Definition 0.1. We call a function $\gamma : [l] \to [k]$ loop of length l if it is a function from integers to integers. Moreover we say a loop is closed if $\gamma(1) = \gamma(l)$.

For example, the loop of $1 \to 4 \to 3 \to 1$ is denoted by $\gamma(1) = 1, \gamma(2) = 4, \gamma(3) = 4, \gamma(4) = 3$.

Let $m_1 + m_2 + ...m_k = n$. Then, $M_n(\mathbb{C}) \cong M_{m_1}(\mathbb{C}) \oplus M_{m_2}(\mathbb{C}) \oplus ... \oplus M_{m_k}(\mathbb{C})$. For $A \in M_n(\mathbb{C})$, we have block representation $A = \left(A_{ij}\right)_{ij}$ where $A_{ij} \in M_{i \times j}(\mathbb{C})$.

Remark. Given a loop γ of length l, we denote $A_\gamma := A_{\gamma(1)\gamma(2)} A_{\gamma(2)\gamma(3)} ... A_{\gamma(l-1)\gamma(l)}$. Note that in this case, let P_i be the standard orthogonal projection to $M_i(\mathbb{C})$. Then, $A_\gamma = P_{\gamma(1)} A P_{\gamma(2)} ... P_{\gamma(l-1)} A P_{\gamma(l)}$.

To start with, given $A, B \in M_n(\mathbb{C})^+$, lets consider the set $I(A, B) = \{\varphi \text{ is cptp} : \varphi(A) = B\}$ first. It is apparent that $I(A, B) = \{\psi \text{ is cp} : \psi(I) = B, \psi^*(I) = A\}$.

To see this, let $\psi(X) = \varphi\left(\sqrt{A} X \sqrt{A}\right)$. Then, since $\langle X, \psi^*(Y)\rangle = \langle \psi(X), Y\rangle = \langle \sqrt{A} X \sqrt{A}, \varphi^*(X)) = \langle X, \sqrt{A}\varphi^*(X)\sqrt{A}\rangle$, we have $\psi^*(X) = \sqrt{A}\varphi^*(X)\sqrt{A}$ and $\varphi^*(I) = A$. Conversely, one can achieve by letting $\varphi(X) = A^{-\frac{1}{2}} \psi(X) A^{-\frac{1}{2}}$.

Denote $\psi(X) = \sum_i V_i X V_i^*$ and let $V_i = W_i \sqrt{D_i} Z_i^*$ be its SVD decomposition. Then, the condition becomes $\sum_i W_i D_i W_i^* = B$, $\sum_i Z_i D_i Z_i^* = A$.

In particular, this is asking for a given set of Hermitian matrices H_i with its spectrum $d^i = (d_1^i, d_2^i, ..., d_n^i)$ can their sum be A or B?

Define Horn polytope

$$\mathfrak{H}_r\left(d^{(1)}, d^{(2)}, ...d^{(r)}\right) = \left\{ \lambda\left(\sum_i W_i D_i W_i^*\right) : W_i \in U(n) \right\}$$

Denote $\boldsymbol{d} = \left(d^{(1)}, ...d^{(r)}\right)$.

Then, $\sum_i W_i D_i W_i^* = B$ is possible if and only if $b = \mathrm{sp}(B) \in \mathfrak{H}_r(\boldsymbol{d})$. Similarly, we have the second condition if and only if $a = \mathrm{sp}(A) \in \mathfrak{H}_r(\boldsymbol{d})$. In conclusion, we have the desired D_i if and only if $h, b \in \mathfrak{H}_r(\boldsymbol{d})$.

To construct all solutions, we also need to consider all the unitaries. Denote

$$\mathcal{D}_r(h, b) = \{\boldsymbol{d} : h, b \in \mathfrak{H}_r(\boldsymbol{d})\}$$

Let $\mathcal{F}_b(\boldsymbol{d}) = \left\{(W_i)_1^r : \sum_i W_i D_i W_i^* = \mathrm{diag}(b)\right\}$ and similarly for $\mathcal{F}_a(\boldsymbol{d})$.

Then, $I(A, B) \cong \mathcal{F}_b(\boldsymbol{d}) \times \mathcal{F}_a(\boldsymbol{d}) / \sim$, where $\sim$ is to admit the uniqueness of svd representation.

In particular every cptp φ with $\varphi(A) = B$ has the form

$$\varphi(X) = \sum_i W_i \sqrt{D_i} Z_i^* A^{-\frac{1}{2}} X A^{-\frac{1}{2}} Z_i \sqrt{D_i} W_i^*$$

where $\boldsymbol{d} \in \mathcal{D}_r(h, b)$ and $(W_i) \in \mathcal{F}_b(\boldsymbol{d}), (Z_i) \in \mathcal{F}_h(\boldsymbol{d})$.

Now, lets consider some trivial case. For the Kraus rank one cptp map, we have the following proposition.

Proposition 0.2. Given $A_1, A_2, B_1, B_2 \in M_n(\mathbb{C})^+$, where $A_1 = H_A D_A H_A^*, B_1 = H_B D_B H_B^*$. The followings are equivalent

1. There exists a cptp map $\varphi : M_n(\mathbb{C}) \to M_n(\mathbb{C})$ with kraus rank one such that $\varphi(A_i) = B_i$.
2. $\left(H_A^* A_i H_A\right)_\gamma = U \left(H_B^* B_i H_B\right)_\gamma U^*$ for some unitary $U \in U(n)$ for some loop γ of length l.
3. $\mathrm{Tr}(w(H_A^* A_1 H_A, H_A^* A_2 H_A)) = \mathrm{Tr}(w(H_B^* B_1 H_B, H_B^* B_2 H_B))$ for every word w.

Proof. We will show $(1) \Rightarrow (2) \Rightarrow (1)$ and $(2) \Leftrightarrow (3)$. Assume there is some kraus rank one cptp map φ such that (1) is valid. Then, let $\psi(X) = H_B \varphi(H_A^* X H_A) H_B^*$ and let $\tilde{A}_i = H_A^* A_i H_A, \tilde{B}_i = H_B^* B_i H_B$. We have $\psi\left(\tilde{A}_i\right) = \tilde{B}_i$. Now, since $\psi \in I\left(\tilde{A}_1, \tilde{B}_1\right)$, ψ has the form

$$\psi(X) = W\sqrt{D} Z D_A^{-\frac{1}{2}} X D_A^{-\frac{1}{2}} Z^* \sqrt{D} W^*$$

For $\boldsymbol{d} \in \mathcal{D}_1(a,b)$, there is some W, Z such that $WDW^* = D_A$ and $ZDZ^* = D_B$ which implies that $D = D_A = D_B$. Moreover, we have $WD = D_A W^* = DW^*$ implies $[D, W] = 0$ and similarly, we have $[Z, D] = 0$. This simplifies $\psi(X) = WZXZ^*W^* = UXU^*$ where $[U, D] = 0$.

Next, assume $D = d_1 I \oplus d_2 I \oplus ... \oplus d_k I \in M_{m_1}(\mathbb{C}) \oplus M_{m_2}(\mathbb{C}) \oplus ... \oplus M_{m_k}(\mathbb{C}) \cong M_n(\mathbb{C})$. Then, since $[U, D] = 0$, U has the form of $U = U_1 \oplus U_2 \oplus ... \oplus U_k$, where $U_i \in U(m_i)$. Then, $\left(\psi\left(\tilde{A}_i\right)\right)_{ab} = U_a \tilde{A}_i U_b^* = U E_a \tilde{A}_i E_b U^* = E_a \tilde{B}_i E_b$, then for any closed loop $\gamma : [l] \to [k]$, we have

$$\left(\tilde{B}_i\right)_\gamma = E_{\gamma(1)} \tilde{B}_{i_1} E_{\gamma(2)} \tilde{B}_{i_2} ... E_{\gamma(l-1)} \tilde{B}_{i_{l-1}} E_{\gamma(l)} = U E_{\gamma(1)} \tilde{A}_{i_1} E_{\gamma(2)} \tilde{A}_{i_2} ... E_{\gamma(l-1)} \tilde{A}_{i_{l-1}} E_{\gamma(1)} U^* = \left(\tilde{A}_i\right)_\gamma$$

Thus, (2) is proved. Now, if (2) holds, it is apparent that by choosing closed loop γ with length 2, we have that $E_a \tilde{A}_i E_b$ simeltaneously unitary equivalent to $E_a \tilde{B}_i E_b$ for any $a, b \in [k]$. Suppose this unitary is U, then we have $\tilde{B}_i = U \tilde{A}_i U^*$. Let $\varphi(X) = H_B U H_A^* X H_A U^* H_B^*$, we have φ be cptp map with $\varphi(A_i) = B_i$.

Note that since $\tilde{A}_i$ and $\tilde{B}_i$ are simultaneously unitary similar, according to Wiegmann's criterion [1], we have that it is equivalent to the (3) condition. □

Bibliography

[1] V. Futorny, R. A. Horn, and V. V. Sergeichuk, "Specht's criterion for systems of linear mappings," *Linear Algebra and its Applications*, 2017.